ENTERPRISE INFRASTRUCTURE PLAN

ABC Startup Solutions

Week 2 Portfolio Project

Bachelor of Science in Information Technology

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PART 1: Company Profile

Company Name: ABC Startup Solutions

Nature of Business: A newly established software development and IT consulting company that builds custom web and mobile applications for small and medium-sized businesses.

Company Vision: To become a trusted regional technology partner known for delivering reliable, innovative, and affordable software solutions to growing businesses.

Company Mission: To empower small and medium enterprises through accessible technology, responsive support, and a culture of continuous learning and improvement among our people.

Office Location (Fictional): 3rd Floor, Meridian Business Center, Brgy. San Isidro, Santa Rosa City, Laguna, Philippines

Organizational Structure

ABC Startup Solutions is led by a Chief Executive Officer (CEO), supported by four department heads who report directly to the CEO: the IT Manager, HR Manager, Finance Manager, and Sales Manager. Each manager oversees their respective team members, who report to them for day-to-day operations.

- CEO
- → IT Manager → IT Staff (Helpdesk, Network, Systems)
- → HR Manager → HR Staff
- → Finance Manager → Finance Staff
- → Sales Manager → Sales Staff

Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

PART 2: Enterprise Hardware Inventory

The table below outlines the hardware required to fully equip ABC Startup Solutions' 20 employees across four departments, with quantities based on role, mobility needs, and shared-resource planning.

Asset ID	Hardware	Qty	Department	Purpose / Justification
HW-001	Desktop Computer	12	IT (3), Finance (5), HR (4)	Primary workstations for office-based staff who need stationary, higher-spec machines for daily productivity, accounting, and development work.
HW-002	Laptop	8	Sales (6), IT (2)	Portable devices for Sales staff who meet clients off-site and for IT staff who need mobility for troubleshooting.
HW-003	Server (Tower/Rack)	1	IT / Server Room	Hosts file sharing, internal applications, domain services, and backup jobs for the whole company.
HW-004	Router	1	IT / Server Room	Manages traffic between the internal network and the internet connection.
HW-005	Switch (24-port managed)	2	IT / Server Room, Sales Floor	Connects all wired devices per floor segment; a second switch is added for future expansion and to avoid a single point of congestion.
HW-006	Printer (Network)	2	Shared - Admin Area, Sales Area	Two units placed strategically so no department walks more than a few desks to print; supports both office and client-facing documents.

Asset ID	Hardware	Qty	Department	Purpose / Justification
HW-007	UPS (Uninterruptible Power Supply)	3	Server Room (1), Networking Rack (1), Shared Office Cluster (1)	Protects the server, networking equipment, and a critical desktop cluster from power interruptions and voltage spikes.
HW-008	Wireless Access Point	2	Office Floor (ceiling-mounted)	Provides Wi-Fi coverage across the single floor for laptops and mobile devices; two units avoid dead zones.
HW-009	NAS Storage (2-Bay)	1	IT / Server Room	Provides shared department storage and a secondary backup target separate from the main server.
HW-010	External Backup Drive	2	IT / Server Room	Offline backup copies for disaster recovery, rotated off-site weekly.
HW-011	Monitor	18	IT, HR, Finance, Sales	Standard monitors for all desktop users and select laptop users who need a larger second screen for productivity.

PART 3: Enterprise Software Inventory

The software stack below covers operating systems, productivity tools, and IT/development utilities needed to operate securely and efficiently.

Software	Version	License	Purpose
Windows 11 Pro	23H2	OEM/Volume License (20 seats)	Primary desktop/laptop operating system; Pro edition needed for domain join, BitLocker, and remote management features.
Ubuntu Server	22.04 LTS	Free / Open Source	Operating system for the on-premise server hosting file sharing, internal apps, and backups; LTS chosen for long-term stability and support.
Microsoft Office	365 Business Standard	Subscription (20 seats)	Word, Excel, and Outlook for documentation, financial records, and company email/communication.
VS Code	1.9x (latest)	Free / Open Source	Code editor for the IT team to develop and maintain internal tools and scripts.
Git	2.4x (latest)	Free / Open Source	Version control for source code and configuration files managed by IT.
GitHub Desktop	3.x (latest)	Free	Simplified GUI for Git so non-CLI-comfortable staff can still contribute to and review documentation/code changes.
VirtualBox	7.x (latest)	Free / Open Source (Personal Use)	Lets IT test software, patches, and configurations in isolated virtual machines before rolling changes out company-wide.
Google Chrome	Latest stable	Free	Standard web browser for all employees; supports web-based business tools like the CRM and cloud storage.
Microsoft Defender	Built-in (Windows 11)	Included with Windows 11 Pro	Baseline antivirus and endpoint protection on every workstation and laptop.
AnyDesk	8.x (latest)	Free (Business tier for IT)	Remote support tool so the Helpdesk Technician can troubleshoot employee devices without walking to each desk.
7-Zip	23.x (latest)	Free / Open Source	File compression and archiving for backups, large file transfers, and email attachments.

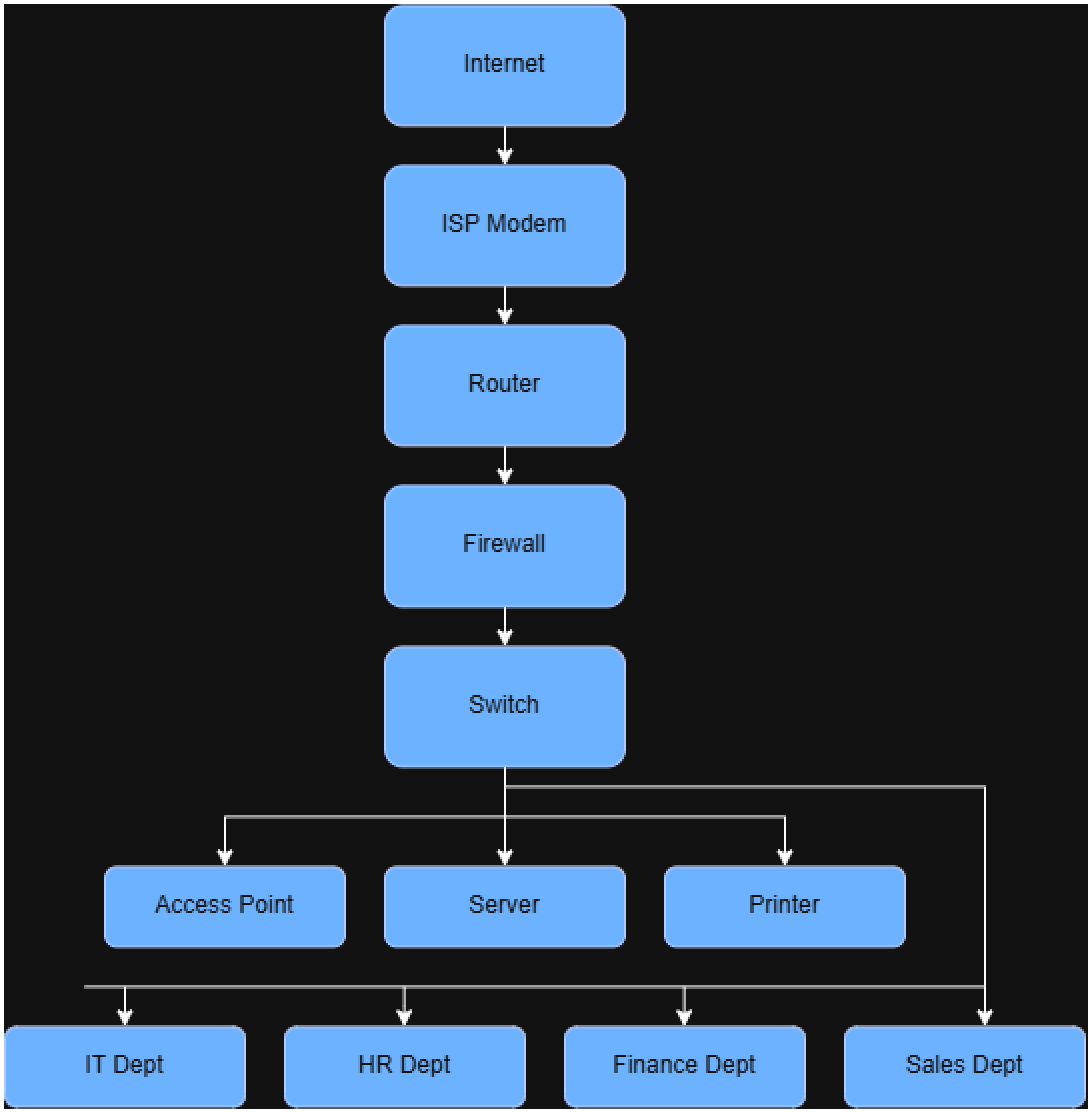
PART 4: Enterprise Network Inventory

The networking equipment below forms the physical and logical backbone connecting all departments to each other and to the internet.

Equipment	Qty	Specification	Purpose
ISP Modem	1	Provided by ISP	Terminates the internet service connection coming into the office.
Router	1	Enterprise-grade (e.g., Cisco/Ubiquiti class)	Routes traffic between the LAN and the internet; handles NAT and basic routing.
Firewall	1	Hardware firewall / UTM appliance	Filters inbound/outbound traffic, blocks intrusions, and enforces network security policy.
Switch	2	24-port Managed Gigabit	Distributes wired connectivity to desktops, printers, and access points; supports VLANs to separate department traffic.
Access Point	2	Ceiling-mount, Wi-Fi 6	Provides wireless coverage across the office floor.

Equipment	Qty	Specification	Purpose
Patch Panel	1	24-port Cat6	Organizes and terminates all structured cabling runs from wall/floor outlets to the switch.
CAT6 Cables	~40	Solid Core, various lengths	Structured cabling connecting each workstation, printer, and access point to the patch panel/switch.
RJ45 Connectors	~80	Cat6-rated	Terminates cable ends at the patch panel and at wall outlets/device ends.

PART 5: Enterprise Network Diagram



PART 6: System Administration Roles

Helpdesk Technician

Responsibilities: First point of contact for end-user IT issues; logs and resolves tickets covering password resets, hardware faults, software installation, and basic connectivity problems; escalates complex issues to Network or Systems Administrators.

Skills: Strong communication and troubleshooting skills, basic knowledge of Windows/Office, ticketing systems, and general hardware diagnostics.

Common Tools: Ticketing systems (e.g., Freshservice, Zendesk), AnyDesk/TeamViewer for remote support, Active Directory Users and Computers.

Certifications: CompTIA A+, ITIL Foundation.

Network Administrator

Responsibilities: Designs, configures, and maintains the company's routers, switches, firewalls, and wireless network; monitors network performance and uptime; manages IP addressing and VLANs.

Skills: Strong understanding of TCP/IP, subnetting, routing/switching, firewall configuration, and network monitoring.

Common Tools: Cisco Packet Tracer, Wireshark, PRTG/Zabbix for monitoring, router/switch/firewall management consoles.

Certifications: CCNA, CompTIA Network+.

Linux System Administrator

Responsibilities: Installs, configures, and maintains the Ubuntu Server used for file sharing, internal apps, and backups; manages users, permissions, and scheduled jobs; applies patches and monitors system health.

Skills: Linux command line proficiency, shell scripting, package/service management, backup and recovery procedures.

Common Tools: Bash, cron, SSH, systemd, rsync.

Certifications: LPIC-1, RHCSA.

Cloud Administrator

Responsibilities: Manages the company's cloud-hosted services (email, backups, potential future cloud servers); handles account provisioning, cost monitoring, and cloud security configuration.

Skills: Familiarity with cloud platforms (AWS/Azure/Google Cloud), identity and access management, cost optimization, cloud security basics.

Common Tools: AWS/Azure/GCP consoles, Terraform, cloud billing dashboards.

Certifications: AWS Certified Cloud Practitioner, Microsoft Azure Fundamentals (AZ-900).

How These Roles Work Together

Within ABC Startup Solutions, these four IT roles work together as a support team instead of working separately. The Helpdesk Technician is usually the first person to handle common problems experienced by employees, such as basic technical issues and user requests. This helps prevent the more specialized IT staff from being overwhelmed with simple concerns.

If the problem is related to the network, such as internet outages, slow connections, or Wi-Fi problems, it can be handled by the Network Administrator. The Network Administrator is responsible for maintaining important networking equipment, including routers, switches, and firewalls, which are needed by the entire company.

The Linux System Administrator is responsible for managing the company's on-premise servers. This includes making sure that file sharing, internal applications, and backups continue to work properly. They may also coordinate with the Network Administrator when there are problems related to server connectivity or network performance.

As the company grows and starts using more cloud-based services, the Cloud Administrator manages these services and helps ensure that employees can access them securely. The Cloud Administrator may also work with the Network Administrator and Linux System Administrator to make sure that the company's on-premise systems and cloud services can work together properly.

Overall, these four roles create a complete IT support structure for the company. The Helpdesk Technician provides first-level support, the Network Administrator manages the network, the Linux System Administrator handles on-premise servers, and the Cloud Administrator manages cloud-based services. By working together, they can help keep ABC Startup Solutions' IT infrastructure organized, reliable, and ready to support the company's growth.

PART 7: Infrastructure Recommendations

Internet Provider: A business-grade fiber connection (e.g., PLDT Business Fibr or Converge ICT Business) with at least 100 Mbps symmetrical bandwidth and a Service Level Agreement (SLA) guaranteeing uptime, since a 20-person company relying on cloud tools and client communication cannot tolerate frequent downtime.

Server Specifications: A single on-premise tower/rack server with a multi-core Xeon or Ryzen processor, 32GB RAM, RAID 1 mirrored storage (for redundancy), and Ubuntu Server LTS as the operating system — sized to comfortably run file sharing, an internal app, and backup jobs for 20 users with headroom for growth.

Backup Strategy: A 3-2-1 backup approach: 3 copies of data, on 2 different media types (server + NAS), with 1 copy stored off-site or in the cloud. Daily incremental backups and a weekly full backup, tested quarterly through a restore drill.

Security Recommendations: A dedicated hardware firewall with intrusion detection, network segmentation via VLANs (separating IT/Finance from Sales/HR), disabling unused ports, and mandatory OS/software patching on a monthly schedule.

Antivirus: Microsoft Defender as the baseline endpoint protection (included with Windows 11 Pro), centrally managed through Microsoft Defender for Business or an equivalent console so the IT team can monitor threats across all 20 machines from one dashboard.

Password Policy: Minimum 12-character passwords combining upper/lowercase letters, numbers, and symbols; mandatory change every 90 days; account lockout after 5 failed attempts; multi-factor authentication (MFA) required for email, cloud, and remote access accounts.

Expansion Plan: The current design (managed switches, extra AP, second switch already budgeted) allows for roughly 30–40% headroom in ports and Wi-Fi capacity. As headcount grows beyond 30 employees, the company should plan to add a second server for redundancy/failover, migrate select workloads to the cloud, and consider a dedicated server room with proper cooling.

PART 8: Personal Reflection

Working on this Enterprise Infrastructure Plan helped me understand more clearly what a System Administrator does even before buying equipment or setting up cables. At first, I thought IT planning was mostly about choosing the right hardware. However, this project showed me that planning starts with understanding the company, its departments, number of employees, and how they will use technology every day. Identifying the needs of ABC Startup Solutions and its 20 employees from the IT, HR, Finance, and Sales departments helped me think about what equipment was actually needed and why. This made the project feel more realistic and similar to the tasks I might do in an entry-level IT job.

The most challenging part for me was creating the network diagram and inventory and making sure they matched each other. Listing the networking equipment in a table was easier, but creating a diagram that correctly showed how the router, firewall, switches, access points, and different departments were connected was more difficult than I expected. I had to check the inventory and diagram several times to make sure that nothing was missing or incorrect. Through this, I learned that proper documentation and network design need to be consistent, especially in real IT work.

This project also helped me understand why planning is important before deploying an IT infrastructure. If ABC Startup Solutions bought equipment without properly planning the network and security policies first, it could lead to incompatible equipment, security problems, and difficulties when the company grows. By first understanding the company's needs, preparing an inventory, designing the network topology, and then making recommendations, the company can avoid unnecessary costs and mistakes. It also gives IT Managers and executives a clear plan that they can review before spending money.

Overall, this project helped me think beyond simply completing a school requirement and imagine myself working as a Junior System Administrator. Researching different IT roles, such as Helpdesk, Network Administrator, Linux Administrator, and Cloud Administrator, also helped me understand the different career paths available in IT. It gave me an idea of the skills and certifications I may need to improve in the future. Because of this project, I feel more confident in understanding a business scenario and turning it into a clear and organized technical plan. I believe these skills will also be helpful for my future capstone project, internships, and career in the IT field.